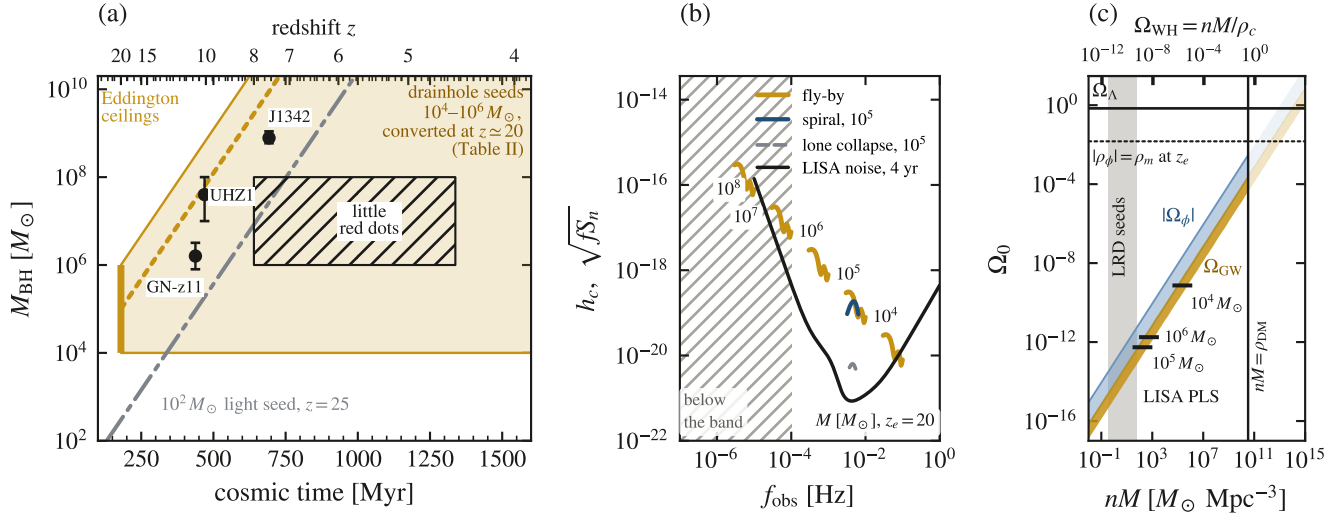

(a)
redshift z
20 15 10 8 7 6 5 4
10^{10}
10^{8}
10^{6}
10^{4}
10^{2}
$M_{\rm BH}\ [M_\odot]$
Eddington ceilings
drainhole seeds 10^4–$10^6\,M_\odot$, converted at $z\simeq 20$ (Table II)
J1342
UHZ1
GN-z11
little red dots
$10^2\,M_\odot$ light seed, $z=25$
250 500 750 1000 1250 1500
cosmic time [Myr]
(b)
10^{-14}
10^{-16}
10^{-18}
10^{-20}
10^{-22}
$h_c,\ \sqrt{fS_n}$
fly-by
spiral, 10^5
lone collapse, 10^5
LISA noise, 4 yr
10^8
10^7
10^6
10^5
10^4
below the band
$M\ [M_\odot]$, $z_e=20$
10^{-6} 10^{-4} 10^{-2} 10^{0}
$f_{\rm obs}$ [Hz]
(c)
$\Omega_{\rm WH}=nM/\rho_c$
10^{-12} 10^{-8} 10^{-4} 10^{0}
10^{0}
10^{-4}
10^{-8}
10^{-12}
10^{-16}
Ω_0
Ω_Λ
$|\rho_\phi|=\rho_m$ at z_e
LRD seeds
$|\Omega_\phi|$
$\Omega_{\rm GW}$
$10^4\,M_\odot$
$10^6\,M_\odot$
$10^5\,M_\odot$
LISA PLS
$nM=\rho_{\rm DM}$
10^{-1} 10^{3} 10^{7} 10^{11} 10^{15}
$nM\ [M_\odot\ {\rm Mpc}^{-3}]$